Conserve mechanical energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Track kinetic and potential energy when nonconservative work is negligible.

SUCCESS CRITERIA - I can...

1. Identify the energy system and assumptions.
2. Keep the total energy consistent between positions.
3. Explain how height and speed trade off.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Lift a 3 kg object by 5 m. Use g=10 N/kg. Find the change in gravitational potential energy.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K_i + U_i = K_f + U_f$; $K = \frac{1}{2}mv^2$; $U = mgh$. Mechanical energy conserved when only conservative interactions transfer energy. Use g = 10 N/kg near the surface.

Worked reasoning: A 1 kg object drops 5 m from rest with g=10 m/s² and no energy loss. It loses 50 J of potential energy and gains 50 J of kinetic energy, giving speed 10 m/s.

Your turn: Draw energy bars at the start and just before arrival. Label the system and neglected losses.

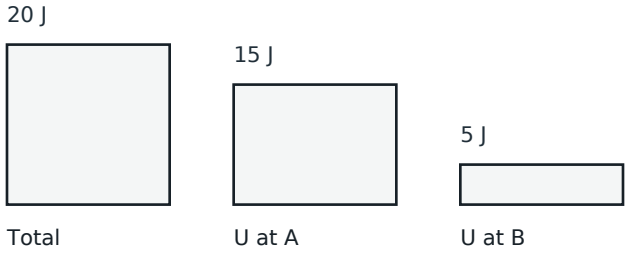
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A system has total mechanical energy 20 J. At one point U=15 J; at another U=5 J. Find K at each point.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Apply the physics

A 1 kg ball is released from rest 2 m above the floor. Use g = 10 N/kg; ignore air resistance. Complete an energy account from release to just before reaching the floor. Find initial U and final K.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A cart loses mechanical energy on a rough track. Does total energy conservation fail? Name the destination to include.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An impactor has 4 times the original mass at the same speed and stopping time. Compare momentum and average stopping-force size.
Use before and after snapshots to organize the momentum account.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

An ideal system has 30 J total. At a point U=12 J, what is K? Use the energy model to show the quantities you compare; label units and any unknowns. Use g = 10 N/kg when needed.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two frictionless ramps start at the same height and end at the same lower height. Compare final speeds and explain what need not be equal. Use the energy model to show the quantities you compare; label units and any unknowns. Use g = 10 N/kg when needed.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3